QR Code and Web-Based Resource Borrowing and Returning System with Availability Tracking, Faculty Authentication, and Email-Based Notifications for MIS Faculty at ICCT Cainta Campus

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Abstract

The borrowing and returning of resources at ICCT Cainta Campus currently relies on manual request letters or faculty ID presentation, which is inefficient, time-consuming, and lacks transparency in tracking availability. Professors often experience delays in borrowing, while MIS faculty struggle with monitoring overdue returns and managing resource quantities. This study proposes a QR code-enabled and web-based automated resource borrowing and returning system to replace the manual request process and streamline transactions. Through QR code scanning, users are directed to the web-based platform where they can register with name, email, phone number, and department information; verify their email address; log in; browse available resources with real-time quantity displays; and submit digital borrowing requests that require administrator approval. Borrowing requests are submitted digitally and require administrator approval, eliminating the need for physical request letters or ID presentation. The system automatically tracks borrowed items, highlights due and overdue returns on the dashboard, and sends email notifications via Brevo API for borrowing approvals, rejections, and return confirmations. The system also provides administrators with tools to manage resources, users, and borrowing transactions through a comprehensive admin *dashboard with visual analytics. By integrating QR code technology, web-based automation, and email notifications, the proposed solution reduces manual workload, accelerates transactions, and enhances transparency in resource management. The expected contribution of this study is an IT-driven solution that modernizes the borrowing and returning process, thereby improving academic support services for MIS faculty at ICCT Cainta Campus.*

Keywords — *QR Code Technology, Web-Based System, Resource Management, Automated Borrowing and Returning, Email Notifications, Information System*

I. INTRODUCTION

A. Background and Motivation

Resource borrowing and returning is an essential support function in academic institutions, particularly for faculty members who rely on shared materials and equipment to deliver instruction effectively. At ICCT Cainta Campus, professors in the MIS department currently request resources either by submitting a formal request letter or by presenting their faculty ID. While this process ensures accountability, it is slow, manual, and prone to delays. MIS faculty also face challenges in monitoring overdue returns and tracking the availability of resources, as there is no centralized system that provides real-time visibility of inventory. These inefficiencies disrupt schedules, reduce productivity, and complicate resource management.

In the field of Information Technology, QR code technology and web-based automation have been widely adopted to streamline workflows, improve accuracy, and enhance transparency. QR codes provide a quick and contactless method for directing users to web platforms, while modern web frameworks such as Laravel provide robust authentication, email integration, role-based access control, and verification modules. This motivates the development of a QR code-enabled web-based system that modernizes the borrowing and returning process for MIS faculty at ICCT Cainta Campus.

B. Problem Statement

The existing borrowing and returning process at ICCT Cainta Campus is inefficient and outdated. Professors must either submit a request letter or present their faculty ID to borrow resources, which consumes time and lacks transparency. MIS faculty struggle to monitor overdue returns and resource availability, as there is no centralized dashboard to track inventory in real time. Furthermore, there is no system for automated email notifications when borrowing requests are approved, rejected, or when items are due for return. These limitations result in delays, poor accountability, and reduced efficiency in resource management.

C. Objectives of the Study

The general objective of this study is to design and develop a QR code-enabled web-based system that automates the borrowing and returning of resources, replacing the current manual process of request letters and faculty ID presentation. Specifically, the study intends to achieve the following objectives:

1. Implement QR code scanning that directs professors to the web-based platform for resource transactions.
2. Provide a web-based dashboard where faculty can view available resources and their

quantities in real time.

3. Allow professors to register with their name, email, phone number, and department, and verify their identity through email verification.
4. Allow professors to submit borrowing requests digitally, eliminating the need for request letters or physical ID presentation.
5. Enable administrators to approve or reject borrowing requests through a dedicated admin interface.
6. Implement automated email notifications via Brevo API to inform users of approval, rejection, and return confirmation status.
7. Enable administrators to monitor active borrows, due and overdue returns, generate transaction reports, and manage resource inventory through the dashboard.
8. Implement role-based access control with faculty and admin roles, institutional email registration with email verification, and a password recovery module.

D. Scope of the Study

The scope defines the specific areas, users, and functionalities that the proposed system will cover:

- **Target Users:** The system is specifically designed for the MIS faculty and professors at the ICCT Cainta Campus.
- **QR Code Integration:** QR codes are provided on campus that redirect professors to the centralized web-based platform for registration, authentication, and resource transactions.
- **Web-Based Platform:** The system is a fully web-based application accessible via standard browsers, featuring role-based access control for administrators and faculty members.
- **User Registration and Profile Management:** Users register using their full name, email address, phone number, and department information. The system supports profile updates including password changes.
- **Email Verification:** New users must verify their email address by clicking a verification link sent by the system via the Brevo API before they can fully access the platform's borrowing features.
- **Password Recovery:** The system includes a forgot password module that sends a password reset link to the user's registered email address via the Brevo API.
- **Real-Time Resource Management:** The system provides real-time tracking and display of available resources and their exact quantities, with visual indicators for available, in-use, and unavailable items.
- **Digital Workflow Submission:** Professors can submit borrowing requests digitally, effectively replacing the need for manual request letters. The system utilizes the professor's email for secure authentication and to facilitate official transaction logs.
- **Administrative Oversight:** Administrators are equipped with tools to approve or reject borrowing requests, manage resources (add, edit, delete), manage user accounts (edit, delete), and monitor both due and overdue returns through a comprehensive dashboard with analytics charts (weekly borrow trends, most used resources ranking, resource status bars).

- **Email-Based Notifications:** The system sends automated email notifications using Brevo API to inform users when their borrow request is approved or rejected, and when items are marked as returned. Email notifications are also sent for email verification upon registration and for password recovery. The sender name is displayed as "ICCT Cainta Campus MIS" with a no-reply email address.
- **In-System Notifications:** The system stores notification records in the database with read/unread status, accessible within the dashboard for users to review.
- **Technical Platform:** The solution is built with Laravel 11 (PHP), utilizing MySQL for data storage, Apache as the web server, Tailwind CSS for responsive design, Alpine.js for interactive components, and Brevo (Sendinblue) API for transactional email delivery. QR codes are generated as standard PNG images linking to the platform URL. Laravel Sanctum handles API token authentication.

E. Limitations of the Study

The limitations identify the boundaries and external factors that the system cannot control or will not address:

- **Connectivity Requirements:** The system is entirely dependent on internet connectivity to function and synchronize data in real time.
- **Account Credential Dependency:** The system's password recovery and notification features are dependent on the user's access to their registered email address. If a user loses access to their email account, an administrative manual override is required for account recovery.
- **Hardware Dependency:** Users must have access to a QR code scanning device (such as a smartphone or web-enabled device) and a device with a web browser to access the platform.
- **Maintenance Boundaries:** The scope excludes the physical maintenance of resources; the system only tracks their digital status and availability.
- **Infrastructure Constraints:** The system may face challenges in environments with bandwidth limitations or specific device compatibility issues.
- **Geographic and Departmental Focus:** The system is deployed at the ICCT Cainta Campus and is currently confined to the MIS faculty, without addressing scalability for other departments or locations.
- **Data Integrity:** The system relies on the honesty of the borrower to report the actual condition of the resource upon return, as there is no automated hardware sensor to detect physical damage.
- **Customization:** The system may not fully adapt to unique departmental workflows without modifications.
- **Email Sending Limit:** The email notification feature relies on Brevo's free plan, which limits transactional email sending to 300 emails per day.
- **Account Deletion Requirement:** If a user needs to re-register with the same email address, their previous account must first be deleted from the system, as duplicate emails are not permitted.

II. RELATED WORKS

The continuous advancement of information technology has reshaped how academic institutions manage student identification, attendance monitoring, and resource utilization. Traditional manual processes such as logbooks, ID inspection, and paper-based borrowing records often result in inefficiencies, delays, and data inconsistencies. Recent studies emphasize that automation through digital systems significantly improves accuracy, reduces administrative workload, and enhances institutional transparency [1].

QR code technology has become one of the most practical tools for implementing automated monitoring systems in educational environments. According to Roy [1], QR-based systems offer a low-cost yet reliable alternative to manual tracking. These systems reduce human error and streamline the process of capturing data. Similarly, Masalha and Hirzallah [2] demonstrated that QR code scanning enables faster access to digital platforms while maintaining data accuracy and accessibility for administrators.

In addition to access and monitoring, QR codes have been applied in broader academic resource management contexts. Rahman et al. [3] explain that QR technology can be effectively used for tracking borrowed materials such as laboratory tools and library books. Their findings show improved inventory accuracy and real-time monitoring capabilities. By digitizing borrowing records, institutions gain structured data that can support planning and decision-making processes.

The effectiveness of QR and web-based systems also depends on system design and quality standards. Liew and Tan [4] highlighted the importance of authentication accuracy and fraud prevention in web-based applications. Meanwhile, Sharma et al. [5] stressed that academic information systems must comply with quality models such as ISO/IEC 25010 to ensure usability, reliability, performance efficiency, and compatibility across devices. Without proper system design considerations, digital platforms may face adoption challenges among users.

Several studies have explored integrated systems that combine QR-based access with resource management functions. Singh et al. [11] developed a QR-based access control and resource tracking system that reduced manual processing time and minimized record errors. Hariyanto et al. [14] implemented a QR student management system featuring login verification and asset checkout workflows. Mahmood et al. [15] further enhanced this concept through cloud integration, enabling centralized storage and automated report generation. These studies confirm that integrating QR-based identification and borrowing functions improves efficiency and strengthens institutional monitoring.

Within the Philippine context, QR-based systems have been widely adopted for attendance monitoring and ID validation. Local studies report improvements in efficiency, user acceptability, and administrative workload reduction [6], [7]. Research on QR and SMS-based monitoring also highlights enhanced punctuality and accountability among students [8], [10]. Additionally, digital ID validation systems have demonstrated faster verification procedures and improved security compared to traditional manual inspection methods [9]. A QR-integrated library circulation system implemented in a local campus further showed improved transaction speed and record accuracy [20].

Email-based notification systems have also been shown to improve user engagement and timely responses in academic platforms. The use of transactional email APIs such as Brevo (formerly Sendinblue) enables cost-effective delivery of verification emails, approval notices, and reminders. Studies on notification systems in academic environments indicate that automated email alerts significantly reduce missed communications and improve overall response times [16]–[19].

Despite the proven effectiveness of QR and web-based systems, current implementations reveal several limitations. Many focus on isolated functions such as attendance tracking or library circulation, without providing a comprehensive solution that integrates QR code access, authentication, resource availability tracking, email notifications, and borrowing management on one platform. Foreign-developed systems often assume robust technological infrastructure, which may not align with local institutional constraints such as bandwidth and device compatibility. Local studies, while effective in specific contexts, lack scalability across multiple departments and do not fully address overdue monitoring, automated notifications, or centralized reporting.

RESEARCH GAP

These gaps highlight the need for a centralized QR code-enabled web-based faculty authentication and resource borrowing management system with email-based notifications, tailored to the operational environment of ICCT Cainta Campus. By integrating QR code access, authentication, real-time availability tracking, digital borrowing requests, email notifications, and overdue monitoring into a single web-based platform built on the Laravel framework, the proposed system aims to enhance efficiency, accountability, and institutional digital transformation for MIS faculty.

III. METHODOLOGY

A. Research Design

This study adopts a developmental research design, which focuses on the systematic analysis, design, and development of an information system to address identified inefficiencies. Developmental research is appropriate because the study aims to create a new IT solution based on existing problems in the borrowing and returning process at ICCT Cainta Campus. The methodology emphasizes both conceptualization and actual implementation, ensuring that the proposed solution is grounded in user needs and technical feasibility.

B. Development Methodology

The system was developed using the Laravel 11 framework, following the Model-View-Controller (MVC) architectural pattern. The development process consisted of the following phases:

1. Requirements Analysis: The existing manual borrowing process was analyzed to identify pain points, including the time-consuming nature of request letters, lack of real-time availability data, and absence of automated notifications.

2. System Architecture Design: The system architecture was designed with the following layers:

- **Presentation Layer:** Blade templates with Tailwind CSS frontend and Alpine.js for reactive components, served via Apache web server. QR codes are generated as PNG images linking to the application URL.
- **Application Layer:** Laravel 11 controllers and services handling business logic, authentication, authorization, email dispatching via Brevo API, and email verification.
- **Data Layer:** MySQL relational database storing users (with email, phone, department, role, verification status), resources, borrows, and notifications.

- **Email Layer:** Brevo (Sendinblue) REST API via Symfony Mailer transport for transactional email delivery, with sender name configured as "ICCT Cainta Campus MIS" and reply-to set to noreply@icct-mis.classapparelph.com.

3. QR Code Integration: QR codes are generated using the qrencode library and are placed within the ICCT Cainta Campus for easy access. Scanning the QR code with any standard QR reader application directs the user to the MIS platform URL. This provides a quick and contactless method for faculty members to access the system without needing to manually type the website address.

4. Database Design: The database schema includes the following primary tables:

- **users** — Stores faculty and administrator accounts with fields: id, name, email, phone, email_verified_at, password, remember_token, role (faculty or admin), department, created_at, and updated_at.
- **resources** — Stores resource inventory with fields: id, name, description, quantity, is_available, created_at, and updated_at.
- **borrows** — Records borrowing transactions with fields: id, user_id, resource_id, quantity, status (pending, approved, rejected, borrowed, returned), borrowed_at, due_at, returned_at, and notes.
- **notifications** — Stores in-system notifications with fields: id, user_id, title, message, is_read (boolean), created_at, and updated_at.

5. Core Features Implementation:

User Authentication and Registration: New users register using their full name, email address, phone number, department, and password. Upon registration, the system sends an email verification link to the user's registered email via the Brevo API. The user must click this link to verify their email address before they can fully utilize borrowing features.

Password Recovery: The system includes a forgot password module accessible from the login page. Users enter their registered email address, and the system sends a password reset link via the Brevo API. The reset link allows users to set a new password through a secure token-based form.

Role-Based Access Control: The system implements two roles: 'faculty' and 'admin'. Faculty users can browse resources, view availability, submit borrowing requests, and view their borrowing history. Administrators have additional capabilities including resource management (add, edit, delete), user management (edit details, delete accounts), borrowing request approval/rejection, and access to system-wide analytics.

Resource Management: Administrators can manage the resource inventory through a dedicated interface with create, read, update, and delete (CRUD) operations. Each resource has a name, description, total quantity, and real-time availability calculated based on active borrows. Faculty users can browse all resources with visual indicators showing available vs. in-use quantities.

Borrowing Workflow: The borrowing process follows a digital request-approval workflow:

1. Faculty selects a resource and submits a digital borrowing request specifying quantity and due date.

2. The request is stored with 'pending' status and appears in the admin dashboard's pending requests section.
3. An administrator reviews the request and either approves or rejects it with optional notes.
4. Upon approval or rejection, the system creates an in-system notification and sends an email notification to the faculty member's registered email via the Brevo API.
5. When the faculty member returns the resource, the administrator marks it as returned in the system, triggering a return confirmation notification.

Dashboard and Analytics: The admin dashboard displays key metrics including total resources, available items, active borrows, pending requests, and overdue items. A weekly transaction chart provides visual trend analysis of borrow activities over time. A "most used resources" ranking section identifies high-demand items. Resource status bars visually depict the proportion of available, in-use, and unavailable resources. The faculty dashboard shows personal borrowing statistics (total borrows, active items, overdue items) and quick access buttons for browsing available resources.

Notification System: The system implements a dual notification approach:

- **In-system notifications** stored in the database with title, message, and read/unread status, displayed within the dashboard header with an unread count badge.
- **Email notifications** sent via Brevo API for the following events: email verification upon registration, password reset links, borrow request approval, borrow request rejection, and return confirmation. The sender name is configured as "ICCT Cainta Campus MIS" with a no-reply email address.

6. Technology Stack:

- Laravel 11 (PHP framework with MVC architecture)
- MySQL (relational database)
- Apache (web server)
- Tailwind CSS (responsive frontend styling)
- Alpine.js (lightweight JavaScript interactivity)
- Brevo API / Sendinblue (transactional email delivery via Symfony Mailer transport)
- QRencode library (QR code generation)
- Laravel Sanctum (API token authentication)

C. System Architecture

The system follows a client-server architecture where QR codes serve as quick-access entry points to the web platform. Scanning a QR code with a mobile device redirects the user to the platform URL via the default web browser. The Apache web server serves Blade-rendered HTML pages to the client browser. All business logic is executed on the server side through Laravel controllers. The MySQL database stores all persistent data, while the Brevo API handles outbound email delivery through HTTPS REST API calls rather than traditional SMTP, ensuring reliable email delivery even in environments where SMTP ports are restricted.

D. Evaluation Plan

To evaluate the effectiveness of the developed system, the following evaluation criteria will be applied:

- **Functionality Testing:** Verification that all core features — QR code access, user registration, email verification, login, resource browsing, borrow request submission, admin approval/rejection, email notifications, password recovery, and dashboard analytics — operate correctly.
- **User Acceptance:** Feedback from MIS faculty and administrators through structured questionnaires covering usability, efficiency improvements, and satisfaction with the digital workflow compared to the manual process.
- **Performance:** Assessment of page load times, QR code scan-to-load latency, email delivery latency, and system responsiveness under normal usage conditions.

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